

The Energy Cost of Privacy in Federated Learning

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Abstract—Differential privacy in federated learning is evaluated along two axes, privacy and utility. On battery-powered edge devices a third decides whether a configuration is deployable at all, and it is not reported. We measure the energy cost of client-level differential privacy with NVML hardware counters differenced around each federated round, sweeping the privacy budget on UCI-HAR, whose clients are the individual study participants, and on CIFAR-10. The cost separates in two. A per-round overhead from per-sample gradients and clipping costs +51% on HAR and +34% on CIFAR-10, carrying no systematic dependence on ϵ ; it decomposes into a power factor of $1.19\times$ that is the same on both datasets to within 0.5% and a duration factor that is not. Everything budget-dependent acts through the number of rounds required, and is the larger term. Measured as energy-to-target-accuracy, privacy is free at 0.40 accuracy on HAR, costs $5.7\times$ at 0.70, and cannot buy 0.85 at any expenditure. Because the cost is paid in joules it inherits the local carbon intensity, an $18.8\times$ range between Algeria’s grid and Norway’s for an identical guarantee. We also report two measurement hazards: running the privacy accountant inside the timed loop reproduces an entire apparent ϵ trend, and calibrating the noise against expected client participation leaves all 29 of our private runs spending more than their stated budget, by up to 22%.

Index Terms—federated learning, differential privacy, energy efficiency, green AI, carbon footprint, edge computing, human activity recognition

I. INTRODUCTION

Differential privacy in federated learning is evaluated almost exclusively along two axes: a budget ϵ is chosen, the resulting accuracy is reported, and the trade-off between them is the result [1], [2].

A third axis is missing. Adding differential privacy changes how much energy training consumes, and on the battery-powered edge devices federated learning exists to serve, energy decides whether a deployment is possible at all. A privacy configuration a phone cannot afford to run is not a point on the privacy–utility curve; it is off the curve. Green AI research has established that energy and carbon deserve first-class measurement [3], [4], but has concentrated on large centralised training. Federated work on the energy of differential privacy has mostly minimised it analytically under a guarantee taken as fixed [5], [6]; where it has been measured on real hardware, it has been at a single privacy setting [7]. The shape of the cost as ϵ varies — which is what a designer needs in order to choose a budget — is what we measure here.

We measure it with NVML hardware counters differenced around each federated round, rather than estimating from

device power ratings, and report three findings a two-axis evaluation cannot produce.

The dominant cost is convergence, not arithmetic. Privacy imposes a per-round overhead from per-sample gradients and clipping — +51% on UCI-HAR, +34% on CIFAR-10 — that carries no systematic dependence on ϵ . It decomposes into a power factor of $1.19\times$, identical on both datasets to within 0.5%, and a duration factor that is not. It is the visible cost and the smaller one. Everything budget-dependent acts through the number of rounds required, which an energy-per-round evaluation cannot see. The two call for opposite responses: a fixed overhead is attacked by optimising the implementation, a convergence penalty by changing the round budget.

The cost grows with the accuracy demanded, until it is unbounded. On HAR, private and non-private training cost the same energy to reach 0.40 accuracy, $1.28\text{--}1.79\times$ to reach 0.50, and $5.7\times$ to reach 0.70. At 0.85, which the non-private model reaches comfortably, no budget we tested arrives at any expenditure: under client-level accounting the extra training that would close the gap is itself what raises the noise.

The same budget carries very different carbon costs. The privacy premium on 1000 HAR runs is $106\text{--}148\text{ gCO}_2\text{eq}$ on Algeria’s gas-dominated grid against $5.7\text{--}7.9\text{ g}$ in Norway. Selecting ϵ is partly a geographic decision, which the privacy–utility framing cannot express.

UCI-HAR [8] is our primary dataset: its clients are the individual volunteers the data came from, so the client boundary is a real person rather than a synthetic partition. CIFAR-10 checks the same effects on a harder benchmark.

One methodological result follows from the measurement itself. Deriving the noise multiplier requires a search over the privacy accountant, and that search costs more for a loose budget than a tight one. Run inside the federated loop, where the natural implementation puts it, its cost lands inside the measurement window and scales with ϵ , taking exactly the shape of a cost of privacy. In one of our sweeps it accounted for the entire observed trend. Any energy study of a privacy mechanism is exposed to this, so we describe it and the controls that remove it.

II. RELATED WORK

A. Measuring the energy of training

Green AI made energy and carbon reportable quantities rather than externalities [3], [4], and a family of tools now estimates them from utilisation traces and grid factors [9].

Comparisons of those tools against instrumented hardware find the estimates diverge substantially from measurement [10], which is why we read NVML counters rather than model power from device ratings. Patterson et al. [11] established that where a job runs changes its emissions by 5–10 $\times$; our geographic result is that finding applied to a parameter, ε , rather than to a schedule. Savazzi et al. [12] give an analytical energy and carbon framework for federated and decentralised learning, but without a privacy mechanism.

B. Energy and differential privacy in federated learning

One line of work treats DP noise as a constraint inside a wireless resource-allocation problem and minimises device energy under it, either by harvesting channel noise as the privacy mechanism [6], [13] or by co-designing compression with the noise budget [5]. Energy there is an optimisation objective evaluated through a convergence bound; it is not measured, and ε is a constraint rather than a swept variable. Closest to us empirically, Schwermer et al. [7] built a physically distributed edge testbed and reported traffic, energy and training time for a CNN on MNIST with differential privacy enabled, and found software estimates deviating by up to 35% from measurement. Their contrast is DP on versus off at one setting; ours is the shape of the cost across the budget. Aroussi et al. [14] schedule ε adaptively on Raspberry Pi and ESP32 nodes and report a 20–25% energy improvement over a fixed-privacy baseline — the gain of a controller, rather than the cost curve a designer would need to choose the budget in the first place.

C. Rounds as the mechanism

That the cost should act through the round count is known analytically. Wei et al. [15] prove that under client-level DP there is an optimal number of aggregations for a given protection level, beyond which convergence degrades, and later derive a discounting method to find it [16]; the same optimum reappears for personalised federated learning and in control formulations that adapt noise and rounds jointly [17]. These results say where to stop. What they do not say is what continuing costs, because the round is an abstract unit in them. Our measurements price it: at $\varepsilon = 0.5$ on CIFAR-10, 60–68% of a run’s joules are spent after the accuracy peak that this literature predicts — and on UCI-HAR no peak clears the measurement noise at all, so the optimum is real but not always resolvable.

D. Our position

We therefore contribute the missing empirical axis: hardware-counter energy swept across ε under whole-run client-level accounting [1], [18], decomposed into a fixed per-round overhead and a convergence penalty, and reported as energy-to-target-accuracy rather than energy per round or energy per run.

III. METHODOLOGY

A. Federated Setup

Both datasets run on Flower [19] with FedAvg, 50% of clients sampled per round for training and 50% for evaluation,

one local epoch (batch 32, SGD at lr 0.01, momentum 0.9), and three seeds per configuration.

UCI-HAR [8] is primary. We keep the published subject-disjoint split: its 21 training participants become the clients, one per person, and the 9 test participants are held out entirely, so central accuracy measures generalisation to people the federation never saw. *Clients correspond to individual study participants*, so the partition is given by who contributed the data rather than by a heterogeneity parameter. Each client holds about 279 training windows. We train on the raw inertial signals (nine channels, 128 samples per 2.56 s window) with a three-block 1D CNN and global average pooling (37k parameters), close enough to the CIFAR-10 model’s 62k that per-round energy stays comparable. Neither model uses batch normalisation, which Opacus [2] rejects.

Local train and validation sets are split *within each activity class*, in file order, discarding the straddling window: HAR windows overlap by 50%, and participants record the six activities in blocks, so neither a random split nor a single ordered cut leaves every client all six classes on both sides.

We train HAR for 100 rounds, the non-private model reaching 95% of its final accuracy only at round 59. The budget is not a free parameter here, since it enters the accounting directly.

CIFAR-10 is secondary: a five-layer CNN (62k parameters) across 20 clients over 30 rounds, partitioned by a Dirichlet distribution over labels at $\alpha = 0.5$ [20].

B. Privacy Accounting

We enforce *client-level local differential privacy*: each client holds its own (ε, δ) budget over the entire run, and the server is untrusted. The budget is accounted over all local SGD steps a client takes across all rounds in which it participates — not per round, and not per local epoch. For a client with n examples, batch B , R rounds, sampling fraction q_c and E local epochs:

$$s_{\text{total}} = R q_c E \lceil n/B \rceil \quad (1)$$

We invert the RDP accountant [21] once, before training, to find the noise multiplier σ giving the target ε at $\delta = 10^{-5}$ after s_{total} steps, and hold it constant. Precomputation is deliberate: in simulation each client is reconstructed every round, so an engine-internal accountant resets each round and reports a per-round ε while the true budget compounds.

Two properties deserve statement. Opacus samples at rate $1/\lceil n/B \rceil$ rather than B/n , and we calibrate at the rate the sampler actually uses. And we claim **no amplification by client subsampling**, which would require secure aggregation or a shuffler; the reported ε is an upper bound.

Because per-client dataset sizes differ, σ differs per client (Table I). The two datasets sit in different regimes: a HAR client holds 279 windows against CIFAR-10’s 2000, so the sampling rate is roughly seven times higher, and with the longer budget HAR’s convergence requires, σ at $\varepsilon = 0.5$ reaches 17.5–20.6 against 2.8–6.1. Under local DP the round budget is therefore not free: extending training to reach convergence raises σ for every round.

C. What the Reported ε Actually Bounds

The step count above uses the *expected* number of rounds a client joins, $R \cdot q_c$. The realised number is a binomial draw around it: at $R = 100$ and $q_c = 0.5$ the expectation is 50 with a standard deviation of 5, so a client sampled 60 times takes 20% more steps than the accountant was paid for. The usual assurance that a reported ε is an upper bound rests on claiming no subsampling amplification, and says nothing about over-participation.

We therefore recompute each client’s spent budget afterwards, from the participation logged every round, partition sizes rebuilt deterministically, and the same accountant. Participation ranged over 33–64 rounds on HAR and 8–22 on CIFAR-10, and **all 29 private runs exceed their stated budget**, by 4.1% to 22.3%; Table I gives the worst realised budget per condition. A study reporting client-level DP in simulation should either report that figure, as we do, or calibrate σ against a participation bound rather than an expectation.

D. Energy Measurement

Energy comes from the NVML counter `nvmlDeviceGetTotalEnergyConsumption` [22], which accumulates millijoules on the device. Differencing it around each round involves no power sampling and no integration and cannot miss a transient: this is measured, not estimated from a TDP model. The window is server-side, bracketing `configure_train` through `aggregate_evaluate`; all clients share one GPU, so a round-level window captures the full cost regardless of actor scheduling, where measuring inside each client would double-count.

We report energy gross and net of an idle baseline, sampled per run after waiting for board power to stabilise, since a card does not reach its floor when a run ends and probing during that tail understates net energy. Baselines are checked against the sweep median (17.93 W on HAR, sd 0.12) and outliers discarded — a control that caught a run probed while the board was still busy at 300 W. Net energy subtracts the baseline over the wall-clock the counter covers, excluding start-up, central evaluation and teardown.

Three seeds per configuration give an energy noise floor of ± 4.89 W (7.80%) on HAR and ± 5.94 W (8.91%) on CIFAR-10, and an accuracy floor — the same statistic on final accuracy — of ± 0.051 and ± 0.031 . No smaller difference is reported as an effect. Both are measured from the across-seed spread rather than assumed, and the ablations are read against floors measured within a cell rather than across a condition, since grouping an ablation grid by ε would make the effect under study part of its own noise estimate.

E. Experimental Controls

Clipping norm $C = 1.0$ for every ε , since tuning it per budget would confound the comparison.

Identical compute across budgets. We verify per round that executed optimiser steps match those the accountant was

TABLE I
ENERGY PER ROUND ON UCI-HAR (100 ROUNDS), MEANS OVER THREE SEEDS. “WORST ε ” IS THE LARGEST BUDGET ANY ONE CLIENT ACTUALLY SPENT, RECOMPUTED FROM LOGGED PARTICIPATION (SECTION III-C). CIFAR-10 σ AT 30 ROUNDS IN BRACKETS

ε	Net (J)	Overhead	Worst ε	σ range
0.5	20.3	+45.4%	0.570	17.50–20.62 [2.77–6.09]
1	20.0	+43.4%	1.122	9.22–10.94 [1.59–3.32]
2	20.7	+48.3%	2.285	4.94–5.90 [1.04–1.90]
4	22.7	+62.2%	4.250	2.75–3.25 [0.78–1.20]
8	21.5	+54.0%	9.112	1.64–1.90 [0.62–0.85]
∞	14.0	—	—	—

calibrated for: on HAR all six conditions execute 89 steps per round, and the counts agree exactly.

The accountant is excluded from the measurement window. Its search costs 0.086 s at $\varepsilon = 0.5$ against 0.693 s at $\varepsilon = 8$, so run inside the loop it scales with ε inside the measured window. In our first clean sweep it reproduced the entire observed trend (predicted +18.2 s at $\varepsilon = 8$ over $\varepsilon = 0.5$; measured +19.7 s) while step counts stayed flat. All σ values are derived before measurement begins.

Randomised run order. Sweeping ε in a fixed order leaves execution order and ε correlated at $\rho = -0.99$, so drift over a sweep is inseparable from a privacy effect. Randomising reduces this to $\rho = -0.20$.

Determinism deliberately not enforced. Enabling deterministic kernels would change measured energy, i.e. measure a configuration nobody deploys. Reproducibility comes from averaging over seeds.

F. Two Choices Worth Stating

The workload does not saturate the device (60–65 W on a 360 W card), which is why net figures accompany gross ones. Per-channel standardisation constants are computed from the HAR training split and are global across clients, so a federated deployment would have to agree them without pooling raw data. And the round budget is set from the non-private convergence curve, so private conditions inherit a longer budget than any of them needs and a correspondingly larger σ ; their accuracies are a lower bound on what a per- ε tuned budget would give. Section V-F collects what these limits mean for the claims.

IV. RESULTS

Every figure is a mean over three seeds, bounded by the two floors of Section III-D. The HAR sweep passes every gate: 18/18 runs, wall-clock flat to 0.3% across the private conditions against 38.0 s without privacy, and 89 optimiser steps per round throughout. On CIFAR-10 one $\varepsilon = 4$ run was measured a month after the others and is excluded for breaking the flatness gate; including it moves the mean overhead from +34% to +36%, inside the floor either way, so that condition is reported on two seeds.

A. The Per-Round Cost Is Power Times Time

Net of idle, private training costs 21.1 J per round on HAR against 14.0 J, and 107.9 J against 80.4 J on CIFAR-10 —

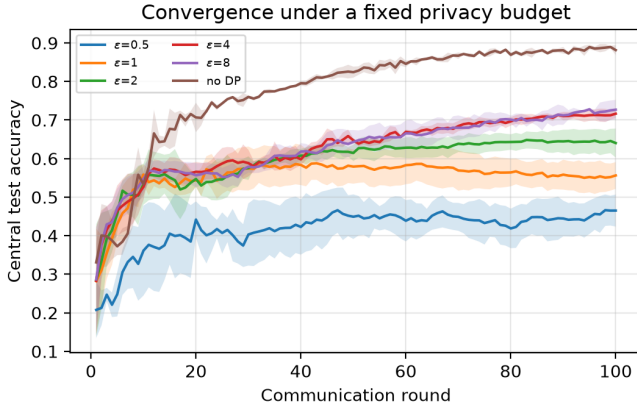


Fig. 1. Central test accuracy against communication round on UCI-HAR over 100 rounds, means of three seeds with ± 1 SD bands. No stopping marker is drawn because none is resolvable here: the largest peak-to-final drop is 0.032 at $\epsilon = 1$ against a ± 0.042 across-seed spread. What the figure shows instead is the quality ceiling — the non-private model reaches 0.88 while $\epsilon = 0.5$ plateaus near 0.47, and further training cannot close that gap because the training itself spends budget.

+51% and +34% (Table I). Gross board power runs 59.3–65.8 W under privacy against 54.5 W without, and 41.6–47.2 W against 36.8 W net of idle. Private rounds draw more power *and* last longer, and the two factors behave differently: the HAR energy ratio $1.507\times$ is $1.190\times$ in power and $1.267\times$ in duration, the CIFAR-10 ratio $1.354\times$ is $1.196\times$ and $1.132\times$. The power factor is the same to within 0.5% across two datasets with different models and a $4\times$ difference in model size — the arithmetic of per-sample gradients and clipping, apparently a hardware constant — while the time factor is not.

Neither orders with ϵ . The overhead runs +45.4%, +43.4%, +48.3%, +62.2%, +54.0% on HAR and +36.0%, +38.6%, +30.7%, +30.6%, +35.6% on CIFAR-10, non-monotone in both. On CIFAR-10 the spread across budgets is 6.0% of the mean, inside the 8.91% floor, so the per-round cost is ϵ -independent within resolution; on HAR it is 12.5% against 7.80%, not resolvably flat, but being non-monotone it is scatter above our resolution rather than a budget effect. Either way the per-round cost carries no systematic budget dependence, which is what the rest of this section rests on.

B. The Cost Grows with the Accuracy Demanded

Measured against the quality a deployment wants, the cost is not flat at all (Table II, Fig. 1). On HAR, reaching 0.40 accuracy costs 185–357 J net whether or not privacy is on: at this quality privacy is free. At 0.50 the private conditions cost 1.28 – $1.79\times$ the non-private 197 J. At 0.70 only $\epsilon \geq 4$ arrives at all, at 1710 J against 301 J, a factor of 5.7. At 0.85, which the non-private model reaches in 57.0 rounds and 839 J, no budget we tested arrives at any expenditure.

CIFAR-10 shows the same shape at its own scale: 0.20 costs 1.40 – $1.63\times$ the non-private 677 J, and 0.25 is reached only at $\epsilon \geq 2$, at 1.96 – $2.29\times$. Unreachability is the sharpest form

of the cost: above the ceiling a budget implies, the accuracy cannot be bought at all, because the extra training that would close the gap is itself what raises σ .

C. A Stopping Round Appears Only Under a Tight Budget

Under a tight enough budget the accuracy curve peaks and declines, so training past the peak spends energy and loses accuracy at once. Whether a peak is resolvable is a question about noise rather than about the argmax, so we report a stopping round only where the peak-to-final drop exceeds that condition’s across-seed spread of final accuracy.

Two CIFAR-10 budgets qualify. At $\epsilon = 0.5$ the mean curve peaks at 0.176 at round 8 and ends at 0.150 — a drop of 0.026 against a ± 0.010 spread, the three seeds peaking at rounds 8, 8 and 9 — and the rounds after the peak account for 60–68% of the run’s net energy. At $\epsilon = 1$ the drop is 0.044 against ± 0.014 , the seeds peak between rounds 10 and 16, and 42–61% of the energy falls after it. At $\epsilon \geq 2$, and without privacy, no decline clears the spread within 30 rounds.

UCI-HAR produces no resolvable stopping round at any budget: its largest drop is 0.032 at $\epsilon = 1$ against a ± 0.042 spread, and at $\epsilon = 0.5$ its seeds peak at rounds 16, 54 and 65 while the mean curve peaks at round 98 — an argmax wandering over a plateau, not a decline. With σ near 20 the model never climbs far enough to give anything back. The effect is real but conditional, the threshold falling between $\epsilon = 1$ and $\epsilon = 2$.

D. Under a Tight Budget, Tuning Stops Working — On One Task

We vary one factor at a time at $\epsilon \in \{1, \infty\}$ with three seeds per cell, against floors measured within a cell: ± 0.041 on accuracy for HAR, ± 0.024 for CIFAR-10.

On HAR the private model is noise-limited. At $\epsilon = 1$ final accuracy spans 0.037 across one, two and five local epochs and 0.018 across $C \in \{0.5, 1, 2\}$, both inside the floor and non-monotone. **On CIFAR-10 the same two knobs are decisive**, falling monotonically from 0.182 to 0.114 to 0.100 across the epoch axis ($3.4\times$ the floor) and from 0.213 to 0.182 to 0.100 across clipping ($4.7\times$). The identical experiment is inert on one task and dominant on the other.

The standard efficiency lever stops paying. More local work per round is how federated learning ordinarily buys fewer rounds, and without privacy it works: on HAR the energy to reach 0.70 falls from 345 J at one local epoch to 193 J at five. Under $\epsilon = 1$ the target is reached at no epoch count while energy per round rises from 22 J to 49 J, and on CIFAR-10 post-peak waste grows from 1785 J to 7312 J. Local steps enter s_{total} , so extra local work forces a larger σ and cancels the gain it bought.

Clipping is free in compute. C leaves energy per round unchanged (22 J at all three values on HAR, 112–119 J on CIFAR-10), so it acts purely through convergence, in a way final accuracy hides: $C = 2$ peaks at 0.187 at round 7 and ends at 0.100, giving back 46.7% of its peak against 2.7% at $C = 0.5$.

TABLE II

ENERGY TO REACH A TARGET ACCURACY ON UCI-HAR (J NET OF IDLE; ROUNDS IN PARENTHESES). “—” MARKS TARGETS NOT REACHED WITHIN 100 ROUNDS. ‡: 2 OF 3 SEEDS

ε	0.40	0.50	0.70	0.85
0.5	357 (13.0)	334‡ (12.0)	—	—
1	201 (4.7)	258 (7.7)	—	—
2	197 (4.0)	353 (12.0)	—	—
4	210 (4.3)	253 (6.3)	1710 (74.3)	—
8	187 (3.3)	269 (7.7)	1699 (76.3)	—
∞	185 (6.0)	197 (7.0)	301 (15.0)	839 (57.0)

TABLE III

GC₂EQ PER 1000 UCI-HAR TRAINING RUNS. “PRIVACY” IS THE PRIVATE MINUS NON-PRIVATE DIFFERENCE ACROSS ε

Grid	DP	No DP	Privacy
Norway	21	15	6–8
France	46	32	12–17
Algeria	403	281	106–148
India	521	363	137–191

E. Carbon Cost by Grid

Table III scales the measured HAR energy to 1000 runs under published grid carbon intensities, repeating the measured workload rather than extrapolating to other hardware. The cost attributable to privacy runs from 106–148 gCO₂eq on Algeria’s gas-dominated grid to 5.7–7.9 g in Norway, a factor of 18.8 for the same computation and guarantee; CIFAR-10 gives 115–146 g against 6.1–7.8 g, the same premium despite a workload five times larger per round. Selecting a privacy budget is therefore partly a geographic decision, which the privacy–utility framing cannot express.

V. DISCUSSION

A. Two Costs, Two Remedies

The fixed per-round overhead is an implementation cost — per-sample gradients and clipping — and is what a systems contribution can attack, through better kernels or amortised clipping. The convergence penalty is bounded by the privacy accounting rather than by any implementation, and can only be answered through the round budget. An evaluation reporting energy per round sees only the first and concludes that privacy costs a flat half again; measured against the accuracy a deployment needs, the second is several times larger. The third axis therefore does not merely rank configurations, it changes which engineering effort is worth spending.

B. Privacy Sets a Quality Ceiling

The cost of privacy is not a constant multiplier: it is invisible at 0.40 accuracy on HAR, 5.7× at 0.70, and unbounded at 0.85. Past the ceiling a budget implies, the trade is not energy for accuracy but nothing for nothing. The usual formulation — choose ε , then see what utility follows — treats the two as sequential; our results suggest choosing them jointly, since a utility-only evaluation reports a lower number without indicating whether more computation would fix it.

The mechanism is that the round budget is part of the privacy budget. Every round a client participates in consumes budget, so σ is fixed before training by a decision federated learning usually treats as free: over the 100 rounds HAR needs, σ at $\varepsilon = 0.5$ reaches 17.5–20.6. Training to convergence is self-defeating, because the extension is what prevents convergence. A stopping rule follows, and we state it conditionally: it is resolvable on CIFAR-10 below $\varepsilon = 2$ and on HAR nowhere, so what generalises is the direction, not a round number.

C. When the Model Is Noise-Limited, Tuning Is Not the Lever

Our HAR ablations are negative and useful. At $\varepsilon = 1$ final accuracy spans 0.037 across local epochs and 0.018 across the clipping norm, both inside the ± 0.041 floor, so a practitioner responding to a disappointing private model by sweeping hyperparameters will spend considerable energy to land where they started. This does not hold in every regime, and CIFAR-10 shows why: there the same two knobs move accuracy by three to five times its floor, because the private model still retains signal to lose. Increasing local work per round is the standard way to reduce communication in federated learning; under DP it stops paying on HAR and actively harms on CIFAR-10, because local steps enter the budget and force a larger σ .

D. Where the Computation Runs

Measured in joules, the cost inherits the carbon intensity of wherever they are drawn: an 18.8× range between Algeria’s grid and Norway’s for identical computation and an identical guarantee. The point is sharper for federated learning than for centralised training, because the computation sits on whatever grids the participating devices do and no operator chooses them. A deployment cannot relocate the way a data centre can; what it can adjust is how much computation the privacy requirement demands.

E. Measuring Energy Around a Privacy Accountant

Instrumentation attached to a privacy mechanism tends to scale with the privacy parameter, so its cost mimics the effect under study. The accountant’s noise-calibration search is the case we hit: run inside the federated loop it reproduced our entire observed ε trend while executed step counts stayed flat, and nothing in the resulting numbers announces this. We recommend three controls: derive noise multipliers before measurement begins, verify per round that executed and calibrated step counts agree, and randomise run order so that drift over a sweep cannot be mistaken for an ε effect.

F. Threats to Validity

Clients are virtual and share one GPU, so the ratios transfer to real devices but the absolute joules do not; confirming them on physical hardware is the immediate next step. Measurement is GPU-only and communication energy is modelled — for a 0.15–0.24 MB model over tens of rounds we expect it small, but have not shown it. One CIFAR-10 run is excluded for being measured in a different session, leaving that condition on two

seeds. And the budgets we report are the targets, not the worst realised ones (Section III-C): the costs here are those of the *stated* guarantee under the weakest trust assumptions, and the guarantee actually delivered is the weaker one in Table I.

VI. CONCLUSION

We measured what differential privacy costs in energy when added to federated learning, using hardware counters around each round on two datasets, with UCI-HAR partitioned so each client holds one study participant’s data.

The cost divides in two. A per-round overhead — +51% on HAR, +34% on CIFAR-10, with no systematic dependence on ϵ — comes from per-sample gradients and clipping, and decomposes into a power factor of $1.19\times$ that is the same on both datasets and a duration factor that is not. Everything else acts through the number of rounds required, and it is the larger term: reaching 0.70 accuracy on HAR costs $5.7\times$ the non-private energy at $\epsilon = 4$ and is unreachable at $\epsilon \leq 2$ regardless of expenditure. Separating them matters more than quantifying either, since they respond to different interventions.

Three rules follow. Derive the round budget from the privacy budget rather than the task, since rounds consume budget and training to convergence is self-defeating — at $\epsilon = 0.5$ on CIFAR-10, 60–68% of a run’s energy falls after the accuracy peak. Choose the privacy budget and the accuracy requirement together, because a budget implies a ceiling no computation reaches. And when the private model is noise-limited, hyperparameter search is not the lever.

Paid in joules, the cost inherits the carbon intensity of wherever training runs: an $18.8\times$ range between Algeria’s grid and Norway’s for an identical guarantee. A privacy–utility evaluation cannot represent this, and it is awkward for federated learning, where the computation sits on whatever grids the participating devices do.

We also report a measurement hazard. Instrumentation attached to a privacy mechanism scales with the privacy parameter, so its cost mimics the effect under study; deriving noise multipliers before measurement, verifying step counts per round, and randomising run order are cheap controls we recommend for any such study.

Two limits bound the claims: clients are virtual and share one GPU, so the ratios transfer but the absolute joules do not, and our two datasets bound the effects rather than establishing them. Characterising where the boundary between those regimes falls — whether it is set by task difficulty, by the magnitude of the noise, or by their ratio — is the question this work leaves open.

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